

Python Code:

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import pandas as pd
import numpy as np
import io, sys
from fpdf import FPDF

# Load raw dataset
df = pd.read_csv(r"E:\CODING\Py\submission_02_reb_tech_academy\retail-orders-raw.csv")

# Step 1: Capture output
buffer = io.StringIO()
sys.stdout = buffer

print("=" * 70)
print("          DATA QUALITY PROFILING REPORT          ")
print("=" * 70)

# 1. COMPLETENESS CHECK
print("\n[1] COMPLETENESS CHECK")
null_counts = df.isnull().sum()
missing_fields = null_counts[null_counts > 0]
if len(missing_fields) > 0:
    for col, count in missing_fields.items():
        pct = (count / len(df)) * 100
        print(f" \n Field '{col}': {count} missing value(s) ({pct:.1f}%)")
else:
    print(" All mandatory fields are 100% complete.")
print("\n" + "*" * 70)

# 2. UNIQUENESS CHECK
print("\n[2] UNIQUENESS CHECK")
dup_count = df.duplicated(subset=['order_id']).sum()
if dup_count > 0:
    print(f" \n Duplicate order IDs detected: {dup_count} duplicate record(s).")
    print(df[df.duplicated(subset=['order_id'], keep=False)][['order_id', 'order_date', 'customer_segment']])
else:
    print(" All order_id records are unique.")
print("\n" + "*" * 70)

# 3. VALIDITY & FORMAT CHECK
print("\n[3] VALIDITY & RANGE CHECKS")

# Quantity Check
print("\n [3.1] QUANTITY CHECK")
for idx, val in df['quantity'].items():
    try:
        q_val = float(val)
        if q_val <= 0:
            print(f" \n Row {idx} (Order {df.loc[idx, 'order_id']}): Invalid quantity = {val} (Must be > 0)")
    except ValueError:
        print(f" \n Row {idx} (Order {df.loc[idx, 'order_id']}): Non-numeric quantity = '{val}'")
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print("\n" + "-" * 70)

# Discount Check
print("\n [3.2] DISCOUNT CHECK")
for idx, val in df['discount_pct'].items():
    if pd.notnull(val) and (val < 0 or val > 100):
        print(f" \n Row {idx} (Order {df.loc[idx, 'order_id']}): Invalid discount = {val}% (Allowed: 0-100%)")
print("\n" + "-" * 70)

# Date Format Check
print("\n [3.3] DATE FORMAT CHECK")
for idx, val in df['order_date'].items():
    if pd.notnull(val):
        try:
            pd.to_datetime(val, format='%Y-%m-%d', errors='raise')
        except Exception:
            print(f" \n Row {idx} (Order {df.loc[idx, 'order_id']}): Non-ISO date format = '{val}'")

print("\n" + "=" * 70)

# Restore normal printing
sys.stdout = sys.__stdout__
output_text = buffer.getvalue()

# Restore normal printing
sys.stdout = sys.__stdout__
output_text = buffer.getvalue() # now buffer exists, so no error

pdf = FPDF()
pdf.add_page()

# Use the full path if needed
pdf.add_font("DejaVu", "", r"E:\CODING\Py\submission_02_reb_tech_academy\DejaVuSans.ttf")
pdf.set_font("DejaVu", size=10)

# Add code
with open(__file__, "r", encoding="utf-8") as f:
    code_text = f.read()
pdf.multi_cell(0, 5, "Python Code:\n\n" + code_text)

# Add output
pdf.multi_cell(0, 5, "\nOutput:\n\n" + output_text)

pdf.output("data_quality_report.pdf")

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Output:

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                DATA QUALITY PROFILING REPORT
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[1] COMPLETENESS CHECK

Field 'order_date': 1 missing value(s) (8.3%)

Field 'city': 1 missing value(s) (8.3%)

Field 'discount_pct': 1 missing value(s) (8.3%)

[2] UNIQUENESS CHECK

Duplicate order IDs detected: 1 duplicate record(s).

	order_id	order_date	customer_segment
3	RT-1004	2026-01-07	Professional
4	RT-1004	2026-01-07	Professional

[3] VALIDITY & RANGE CHECKS

[3.1] QUANTITY CHECK

Row 6 (Order RT-1006): Invalid quantity = -1 (Must be > 0)

Row 8 (Order RT-1008): Non-numeric quantity = 'two'

[3.2] DISCOUNT CHECK

Row 7 (Order RT-1007): Invalid discount = 105.0% (Allowed: 0-100%)

[3.3] DATE FORMAT CHECK

Row 1 (Order RT-1002): Non-ISO date format = '03/01/2026'

Row 6 (Order RT-1006): Non-ISO date format = '2026-13-10'

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